



LESSON - 19

SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

Date & Time Functions

SQL provides several built-in functions to work with date and time data. Here are some commonly used Date & Time functions in SQL:

One SQL Script attached with all examaples

Attached PDF with 150 + Examples

Current Date & Time Functions

Function	Description
NOW()	Returns current date and time (YYYY-MM-DD HH:MM:SS)
CURDATE()	Returns current date (YYYY-MM-DD)
CURTIME()	Returns current time (HH:MM:SS)
SYSDATE()	Returns the time at which the function executes
UTC_DATE()	Returns current UTC date
UTC_TIME()	Returns current UTC time
UTC_TIMESTAMP()	Returns current UTC date and time
LOCALTIME()	Synonym for NOW()
LOCALTIMESTAMP()	Synonym for NOW()

Date Extraction Functions

Function	Description
YEAR(date)	Extracts year
MONTH(date)	Extracts month
DAY(date)	Extracts day of month
DAYOFMONTH(date)	Extracts day of month
DAYOFWEEK(date)	Returns weekday index (1=Sunday)
DAYOFYEAR(date)	Returns day of the year (1–366)
WEEK(date)	Week number
QUARTER(date)	Quarter of the year (1–4)
HOUR(time)	Extracts hour
MINUTE(time)	Extracts minute
SECOND(time)	Extracts second
MONTHNAME(date)	Month name (e.g., August)
DAYNAME(date)	Day name (e.g., Monday)
EXTRACT(unit FROM d)	Extracts any part of date/time

Date Arithmetic Functions

Function	Description
DATE_ADD(date, INTERVAL expr unit)	Adds interval to date
DATE_SUB(date, INTERVAL expr unit)	Subtracts interval from date
ADDDATE(date, INTERVAL expr unit)	Synonym for DATE_ADD()
SUBDATE(date, INTERVAL expr unit)	Synonym for DATE_SUB()
DATEDIFF(date1, date2)	Returns number of days between dates
TIMESTAMPDIFF(unit, datetime1, datetime2)	Difference in specified unit (e.g., MINUTE)

Formatting & Parsing Functions

Function	Description
<code>DATE_FORMAT(date, format)</code>	Formats date to string
<code>TIME_FORMAT(time, format)</code>	Formats time to string
<code>STR_TO_DATE(str, format)</code>	Converts string to date
<code>UNIX_TIMESTAMP()</code>	Returns current Unix timestamp
<code>UNIX_TIMESTAMP(date)</code>	Converts date to Unix timestamp
<code>FROM_UNIXTIME(timestamp)</code>	Converts Unix timestamp to readable format

Miscellaneous Date/Time Functions

Function	Description
LAST_DAY(date)	Returns last day of the month
MAKEDATE(year, dayofyear)	Creates a date from year and day number
MAKETIME(hour, min, sec)	Returns a time from parts
TIMESTAMP(date, time)	Combines date and time into a timestamp
TO_DAYS(date)	Converts date to number of days since year 0
FROM_DAYS(N)	Converts day number to date
SEC_TO_TIME(seconds)	Converts seconds to time string (HH:MM:SS)
TIME_TO_SEC(time)	Converts time string to total seconds